Real-Time E-Commerce
Data Pipeline & Analytics System

Project Documentation & Stakeholder Brief

Prepared by

Ali Hamdy Rizq

Data Engineer

May 2026

1. What Is This Project?

This project is a complete, production-grade Data Engineering Pipeline built to process and analyze real e-commerce product data. It simulates how large companies like Amazon or Noon handle millions of product records — from the moment data arrives, all the way to business dashboards.

The system was built entirely with open-source tools and runs end-to-end on a local machine, making it ideal for demonstrating real-world data engineering skills.

2. The Problem It Solves

E-commerce platforms generate thousands of product records daily. Without a structured pipeline, this data sits in raw CSV files — unusable for business decisions. This project answers:

- Which product categories generate the most revenue?
- Which products have the best quality-to-price ratio?
- How effective are discounts in driving customer ratings?
- Which sellers have the strongest catalog performance?

3. Architecture — How It Works

The pipeline follows the modern Data Lakehouse pattern used by companies like Uber, Airbnb, and Netflix:

Stage	Component	What It Does
1 — Ingestion	producer.py	Reads the CSV and splits it into 50-row batches, simulating a real-time data stream
2 — Raw Data Lake	Parquet Files	Saves each batch as an immutable compressed file, partitioned by date and hour
3 — Processing	transform.py	Cleans prices, fills missing values, and engineers 9 new analytical features
4 — Data Warehouse	SQLite (Star Schema)	Loads clean data into 4 structured tables optimized for business queries
5 — Analytics	Jupyter Notebook	Generates 6 professional business charts and KPI insights
6 — Orchestration	Apache Airflow DAG	Automates and schedules the entire pipeline to run every hour

4. Key Business Numbers

KPI	Value
Total Products Processed	1,000
Product Categories	97
Average Rating	3.62 / 5.0
Average Discount	53.8%
Average Selling Price	INR 1,706
Total Revenue Potential	INR 17,06,096
Known Sellers	699 out of 1,000
Batches Ingested	20 batches (50 rows each)

5. Database Structure (Star Schema)

The data warehouse uses a Star Schema — the industry standard for analytical databases. It consists of 4 tables:

Table	Type	Records	Description
fact_product_listing	Fact Table	1,000	All metrics: price, rating, discount, popularity score
dim_product	Dimension	1,000	Product details: title, currency, variations
dim_category	Dimension	97	All product categories
dim_seller	Dimension	302	All unique sellers

Why Star Schema? It allows fast SQL queries with simple JOINs — perfect for connecting to Power BI, Tableau, or any BI tool.

6. DB Browser for SQLite — What It Is & Why We Use It

DB Browser for SQLite is a free, open-source visual tool for exploring SQLite databases. Think of it as Excel — but for databases.

What you can do with it:

- Browse all 4 tables and see the actual data rows
- Run SQL queries and see results instantly
- Export query results to CSV for further analysis
- Verify data quality — check for nulls, duplicates, wrong values
- View the database schema (table structure)

How to use it:

- **Step 1:** Open DB Browser for SQLite
- **Step 2:** Click 'Open Database' and navigate to: data\ecommerce_dw.db
- **Step 3:** Click 'Browse Data' tab and select any table to view records
- **Step 4:** Click 'Execute SQL' tab and paste any query from analytics_queries.sql
- **Step 5:** Press F5 or click the Play button to run the query

Example Query to try:

```
SELECT category_slug, COUNT(*) as products,
 ROUND(AVG(rating),2) as avg_rating,
 ROUND(SUM(final_price),0) as revenue
FROM fact_product_listing
GROUP BY category_slug
ORDER BY revenue DESC LIMIT 10;
```

7. How to Run the Pipeline

The pipeline runs in 3 simple commands from the project folder:

Step	Command	Output
1 — Ingest	python ingestion/producer.py	Creates 20 Parquet files in data/raw/
2 — Transform	python processing/transform.py	Creates products_cleaned.parquet in data/processed/
3 — Load	python warehouse/loader.py	Populates 4 tables in ecommerce_dw.db

After running these 3 steps, open analytics.ipynb in VS Code and click 'Run All' to generate all charts.

8. Technology Stack

Tool	Purpose	Why Chosen
Python 3.14	Core language	Industry standard for data engineering
Pandas	Data processing	Fast, production-proven DataFrame library
Apache Parquet	Raw storage format	Columnar, compressed, schema-preserving
SQLite	Data warehouse	Zero-config, portable, SQL-standard
Apache Airflow	Orchestration	Industry-standard pipeline scheduler
Jupyter Notebook	Analytics & visualization	Interactive data exploration
GitHub	Version control	Professional code repository
Apache Kafka (ready)	Real-time streaming	Can be enabled for true streaming

9. Future Enhancements

- Migrate to PostgreSQL or Snowflake for production-scale warehousing
- Enable Apache Kafka for true real-time event streaming
- Add Great Expectations for automated data quality validation
- Build a Power BI or Tableau dashboard connected to the warehouse
- Containerize with Docker Compose for one-command deployment
- Deploy to AWS using S3 + Glue + Redshift

Prepared by Ali Hamdy Rizq | GitHub: github.com/Ali632-del | May 2026